Claudio Camolese

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Deep learning enthusiast.

Experience

2025 **AI Research Student UROP – Politecnico di Torino**:
Research project focused on optimizing inference of a Large Language Model (LLM) reasoning on edge devices.
Skills: *LLM, GenAI, Hugging Face*

2024 – 2025 **AI and Perception engineer – Squadra Corse Driverless**:
Worked on the state estimation node using LiDAR odometry. Written documentation for a reinforcement learning agent integrated in Matlab and Simulink. Competed in the Formula SAE competitions.
Skills: *YOLO, LiDAR, ROS2, RViz, C++, Gazebo*

Education

2026 **Master thesis University of Cambridge**.
Investigating PINNs for reducing sim2real gap in deep reinforcement learning.

2024 – 2026 **Double M.Sc. Degree, specialization in AI**.
Double degree program at Politecnico di Torino, Italy and École nationale supérieure d'informatique et de mathématiques appliquées, France.
Relevant courses: *Robotics, Computer Vision, Natural Language Processing, GPU computing, Generative multimodal AI, Machine learning lab, Deep learning, Mathematics in machine learning, Robot learning, Numerical optimization*.

2021 – 2024 **BSc. Physics engineering**.
Bachelor's degree combining quantum physics, electronics and advanced mathematics,
Relevant courses: *Quantum physics, Advanced methods for physics, Solid state physics, Applied Electromagnetism, Nuclear physics*.
Thesis title: *Simulation of a neutron star using TOV equations*.

Projects

2025 **6D Pose Estimation**.
Development of a deep learning pipeline for estimating the 6D pose of objects from single RGB-D images using CNN and GraphNN.

CNN in CUDA from scratch.
Implement a full CUDA CNN training with only C++ standard libraries.

Generative AI for image generation.
Implement diffusion, flow matching, GAN and VAE models for image generation.

VLM for image description.
Implement VLM to map an input image to a natural language description.

Projects (continued)

- **Physics Informed Neural Network for Heat diffusion**.
  Implement PINN to solve thermal diffusion PDE equation.
- **Semantic segmentation on drone dataset**.
  Implement a Deeplabv3+ architecture for image segmentation.
- **Binary semantic segmentation**.
  Implement an end-to-end image segmentation pipeline using a Hourglass and a UNet.
- **Deep learning for 3D shape recovery**.
  Reconstructs 3D shapes from multiple 2D silhouettes using voxel carving and an implicit 3D MLP.
- **LSTM for music generation**.
  Use of an LSTM network for music generation and for merging music genres.
- **Deep Learning Optimization**.
  Implementation and analysis of numerical optimization algorithms applied to large-scale deep learning problems. Focus on convergence analysis and performance.
- **Augmented Reality**.
  AR application that detects a chessboard, calibrates the camera, estimates the pose and projects a 3D object onto the scene in real time.
- **Deep learning for sentiment classification**.
  Train and evaluate a DistilBERT-based model for sentiment classification.
- **Deep Reinforcement learning Flappy Bird**.
  Developed an AI Agent to play Flappy Bird game in gym environment.
- **Unsupervised learning for geometric shapes**.
  Implemented the Spectral Clustering algorithm using the normalized Laplacian approach, based on the eigen decomposition of graph-related matrices derived from a similarity measure.

Skills

Coding	Python, C++, CUDA, Matlab, Java
Tools	LiDAR, YOLO, OpenCV, Git, Pytorch, ROS2, Gazebo, RViz, W&B, CometML, CMake, Docker, Linux, Spark, Hadoop, MongoDB, MLlib, Graphx, SQL, NoSQL, LabView, Arduino

Awards and Achievements

- **UROP undergraduate researcher open badge**
- **Scholarships**: Awarded with five Manager Italia Scholarships and one Stellantis N.V. scholarships for academic excellence.
- **GenNext entrepreneur**: Selected in the entrepreneurship program for top students
- **OpenCV certificate**, **VLM certificate**, **GNU/Linux program open badge**

Languages

2024	**IELTS C1 English certificate**.
2017	**DELF B1 French certificate**.

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